EMPLOYEE ATTENDANCE DASHBOARD

User Manual

How to install, start and use the software, step by step

Project: Employee Attendance Dashboard (AttendTrack)

Made by: Cecil Baraik, B.Sc. IT

Built with: Spring Boot (Java) + React

Source code: github.com/cecilbaraik19/Web-Development (Java Projects/attendance-dashboard)

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1. What is this software?

AttendTrack (Employee Attendance Dashboard) is a web application that keeps a record of when employees come to work and when they leave. It replaces the paper register or Excel sheet that many offices use.

With this software you can:

- Record check-in and check-out time for every employee.
- See who is present, late, on leave or absent today, on one screen.
- Let employees apply for leave and ask for corrections; managers approve or reject them.
- Stop fake check-ins with a rotating QR code, office location (GPS) and office Wi-Fi rules.
- Handle holidays, shifts (morning / general / night) and overtime.
- Download reports in CSV, Excel and PDF, and send email alerts.
- Use it in **light mode** or **dark mode**.

Who uses it? (3 roles)

Role	Who is this?	What they can do
Admin	HR / owner of the company	Everything: all employees, all departments, user accounts, holidays and shifts, check-in security, notifications and audit log.
Manager	Team lead of one department	Sees only **their own department**: dashboard, attendance, approvals, employees, reports. Also checks in for themselves.
Employee	Normal staff member	Sees only **their own** attendance. Checks in / out, applies for leave, asks for corrections.

> **Remember:** The software opens in a web browser (Chrome, Edge, Firefox). It is not a mobile app, but the pages also work on a phone browser.

2. First-time setup (on Windows)

You do this only once on a new computer. The software has two parts that must both run:

Part	Folder	Address	What it does
Backend	`backend`	http://localhost:8080	Java program. Stores data, checks passwords, applies the rules.
Frontend	`frontend`	http://localhost:5173	The website you see and click.

Step 1: Install the tools

Tool	Version	How to check it is installed (in PowerShell)
Java JDK	17 or newer (you have 21)	`java -version`
Maven	3.9 or newer	`mvn -v`
Node.js	18 or newer	`node -v` and `npm -v`
Git	any	`git --version`

If a command says "*is not recognized*", that tool is not installed or not added to the PATH. Install it and open a new PowerShell window.

Step 2: Get the project

The project is already on your computer at:

```
D:\Web-Development\Java Projects\attendance-dashboard
```

On another computer, download it from GitHub:

```
git clone https://github.com/cecilbaraik19/Web-Development.git
```

Step 3: Start the software (easy way)

Open PowerShell and run:

```
cd "D:\Web-Development\Java Projects\attendance-dashboard"
.\start-dev.ps1
```

Two new windows open: one for the backend and one for the frontend. Wait until the backend window shows **"Started AttendanceApplication"** (about 20-40 seconds the first time).

> **Careful:** If Windows says "*running scripts is disabled*", use this instead:
> `powershell -ExecutionPolicy Bypass -File .\start-dev.ps1`

Step 3 (other way): Start both parts by hand

Window 1 - backend:

```
cd "D:\Web-Development\Java Projects\attendance-dashboard\backend"
mvn spring-boot:run
```

Window 2 - frontend:

```
cd "D:\Web-Development\Java Projects\attendance-dashboard\frontend"
npm install      # only the first time
npm run dev
```

Step 4: Open the website

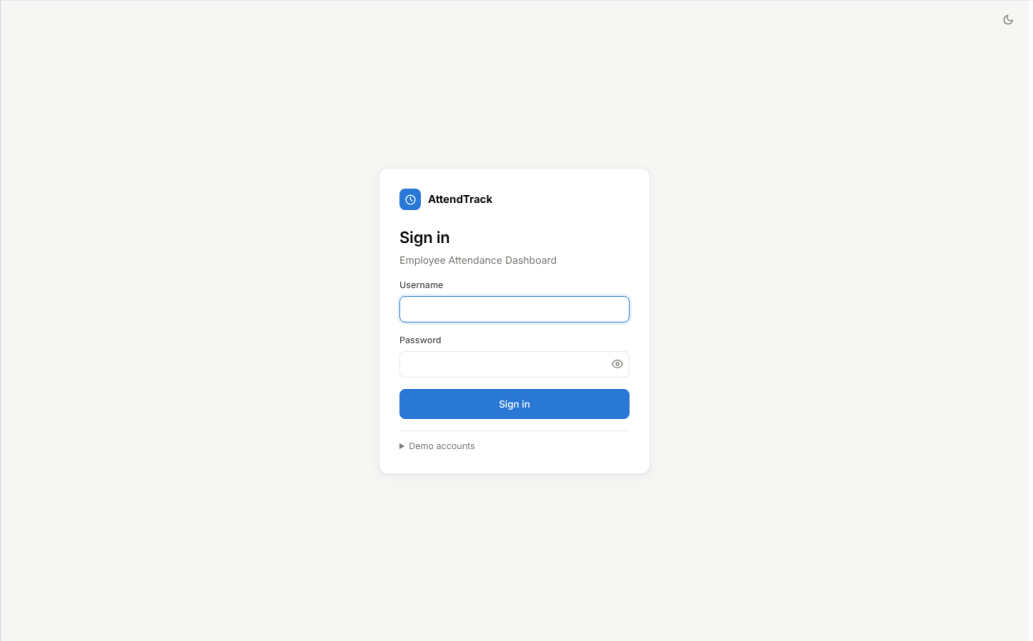
Open **http://localhost:5173** in your browser. You will see the login page.

How to stop the software

Close the two windows, or press **Ctrl + C** inside each window. Your data is saved in the `backend\data` folder, so it is still there next time.

> **Tip:** To start again with fresh demo data, stop the backend and delete the `backend\data` folder. The demo data is created again on the next start.

3. Logging in



The login page

1. Type your **Username** and **Password**.
2. Click **Sign in**.
3. You go to your home page. Admin and Manager see the **Dashboard**. An Employee sees **My Attendance**.

Demo accounts

Username	Password	Role	Person
admin	Admin@123	Admin	-
manager	Manager@123	Manager	Vikram Rao (Engineering)
employee	Employee@123	Employee	Aarav Sharma (Engineering)

Login safety rules

- After **5 wrong passwords**, the account is locked for **15 minutes**.
- A password must have at least **8 characters**, with a capital letter, a small letter and a number. It must not contain the username.
- You stay logged in for **8 hours**. After that you must log in again.

The sidebar

The menu on the left shows only the pages your role may use. At the bottom of the sidebar:

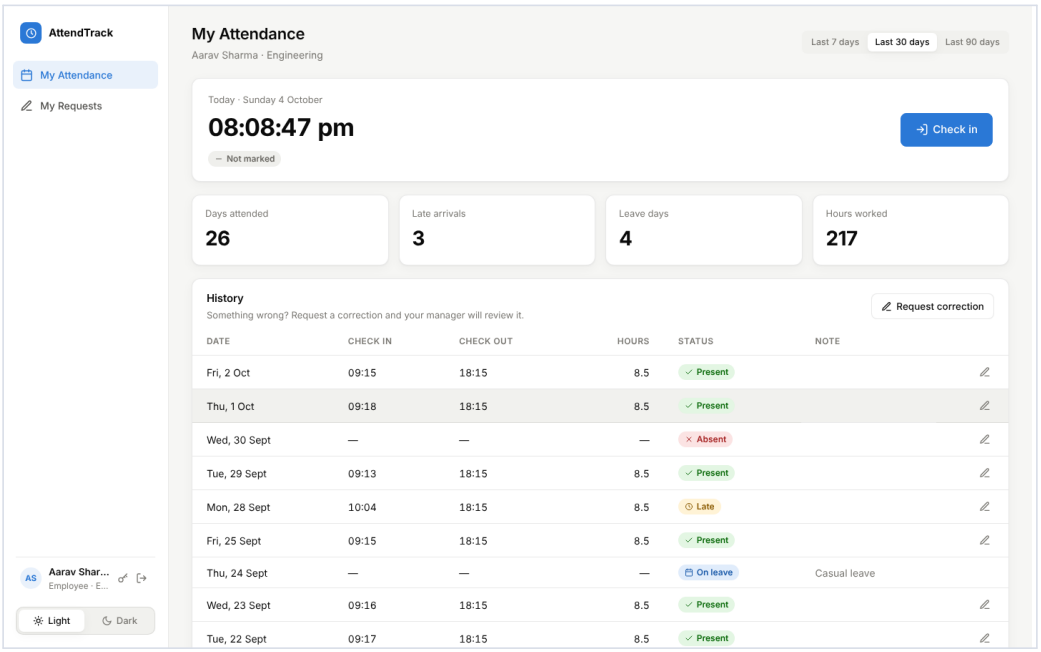
- **Your name and role.**
- **Key icon** - change your password. (Changing it logs you out on all other devices.)
- **Door icon** - sign out.
- **Light / Dark** - switch the colour theme. The software remembers your choice. On the first visit it follows your computer's setting.

> **Careful:** Change the admin password after the first login. "Admin@123" is only for the demo.

4. Employee guide

Log in as **employee / Employee@123** to try this part. An employee sees two pages: **My Attendance** and **My Requests**.

4.1 My Attendance - check in and check out



My Attendance page

1. When you reach the office, open **My Attendance** and click **Check in**. The time is saved.
2. If the admin has turned on the QR rule, a box asks for the **6-digit code**. Type the code shown on the office kiosk screen, or scan the QR code with your phone (see section 7).
3. If the GPS rule is on, the browser asks to use your location. Click **Allow**.
4. When you leave, click **Check out**. Your working hours are calculated.

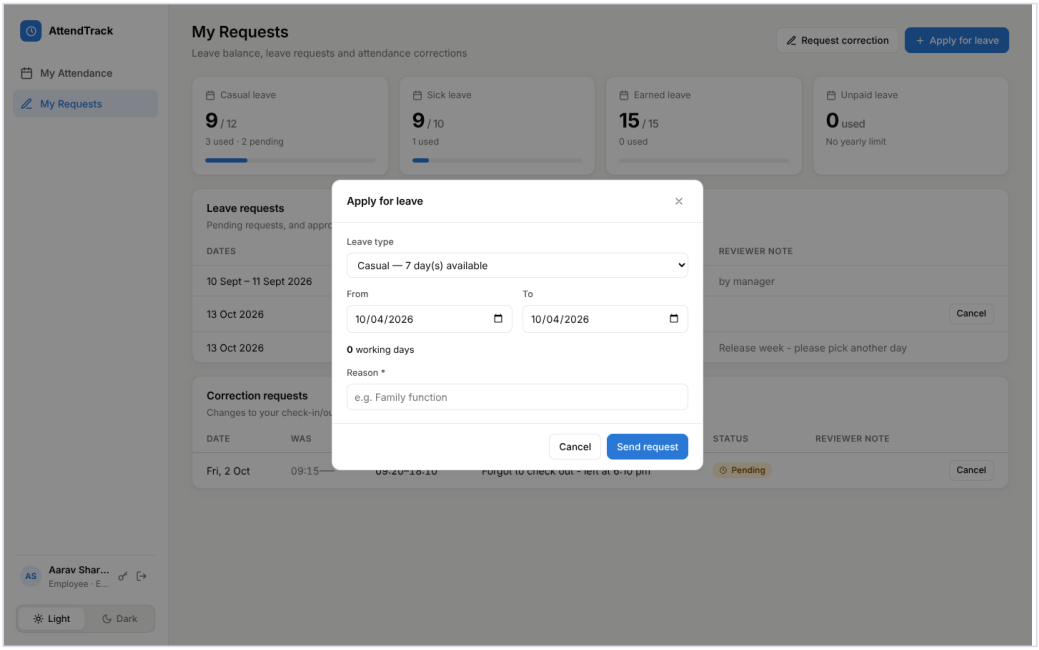
The page also shows:

- **Cards** at the top: days attended, late arrivals, leave days, hours worked, overtime.
- **Period** buttons: last 7, 30 or 90 days.
- **History** table: date, check-in, check-out, hours, status, note.
- **Month view**: a coloured calendar of the month (green = present, yellow = late, blue = leave, red = absent, holidays marked).

What the status words mean

Status	Meaning
Present	Checked in on time (before shift start + 15 minutes grace).
Late	Checked in after the grace time. Example: shift starts 09:30, grace 15 min, so after 09:45 = Late.
Half day	Worked less than 4.5 hours.
On leave	Approved leave for that day.
Absent	Working day, but no check-in and no leave.
Not marked	Today, and you have not checked in yet.

4.2 My Requests - apply for leave



Apply for leave form

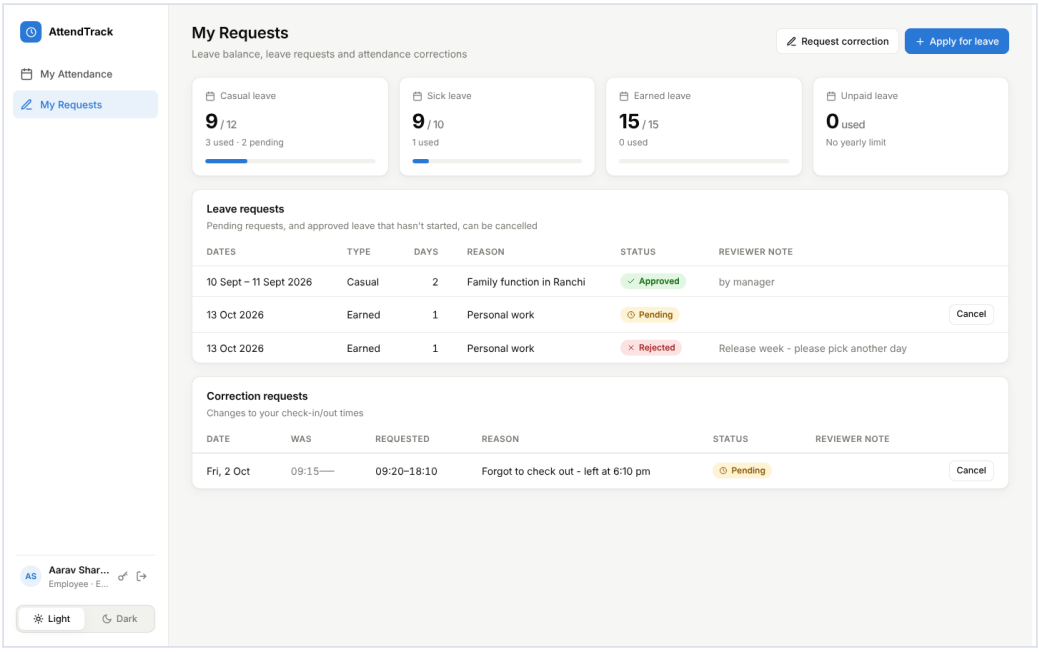
1. Open **My Requests**. At the top you see your leave balance: Casual 12, Sick 10, Earned 15 per year, and Unpaid (no limit).
2. Click **Apply for leave**.
3. Choose the **Leave type**, the **From** and **To** dates and write a **Reason**.
4. The form shows how many **working days** it will use. Weekends and holidays are not counted.
5. Click **Send request**. The status is **Pending** until your manager decides.

When the manager approves, the status becomes **Approved** and those days show as *On leave*. If rejected, you see the manager's reason in **Reviewer note**. You can **Cancel** a pending request, or an approved leave that has not started yet.

4.3 Request a correction

Use this when you forgot to check out, or the time is wrong.

1. Click **Request correction** (on My Requests), or the pencil icon next to a day in the History table.
2. Choose the date, type the correct check-in and check-out time, and write the reason. Example: "*Forgot to check out - left at 6:10 pm*".
3. Send it. When the manager approves, your attendance record is updated automatically.



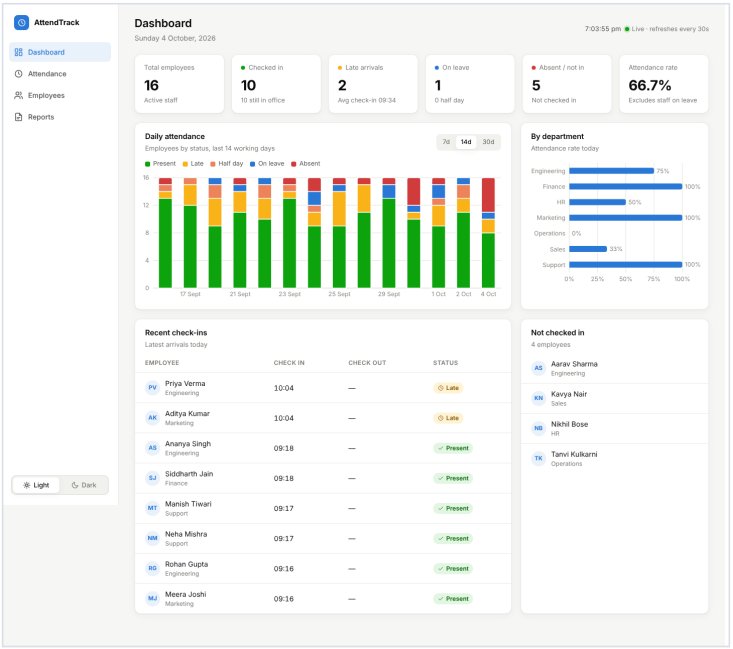
My Requests: leave balance, leave requests and correction requests

My Requests also shows **Upcoming holidays**, so you know which days are off.

5. Manager guide

Log in as **manager / Manager@123**. A manager uses the same pages as the admin, but sees only **their own department** (for the demo: Engineering). A manager also has **My Attendance** and **My Requests** for themselves.

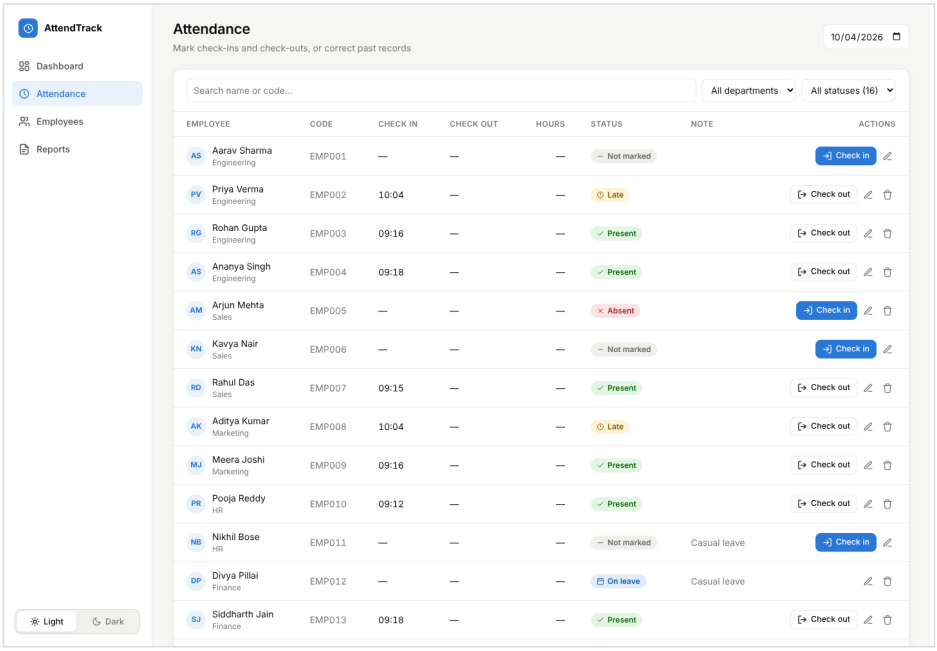
5.1 Dashboard



Dashboard (light mode)

- **Cards**: total employees, checked in, late arrivals, on leave, absent / not in, attendance rate.
- **Daily attendance chart**: present / late / leave / absent for the last 7, 14 or 30 days.
- **By department**: attendance rate of each department.
- **Recent check-ins** and **Not checked in** lists.
- The dashboard refreshes by itself every 30 seconds.

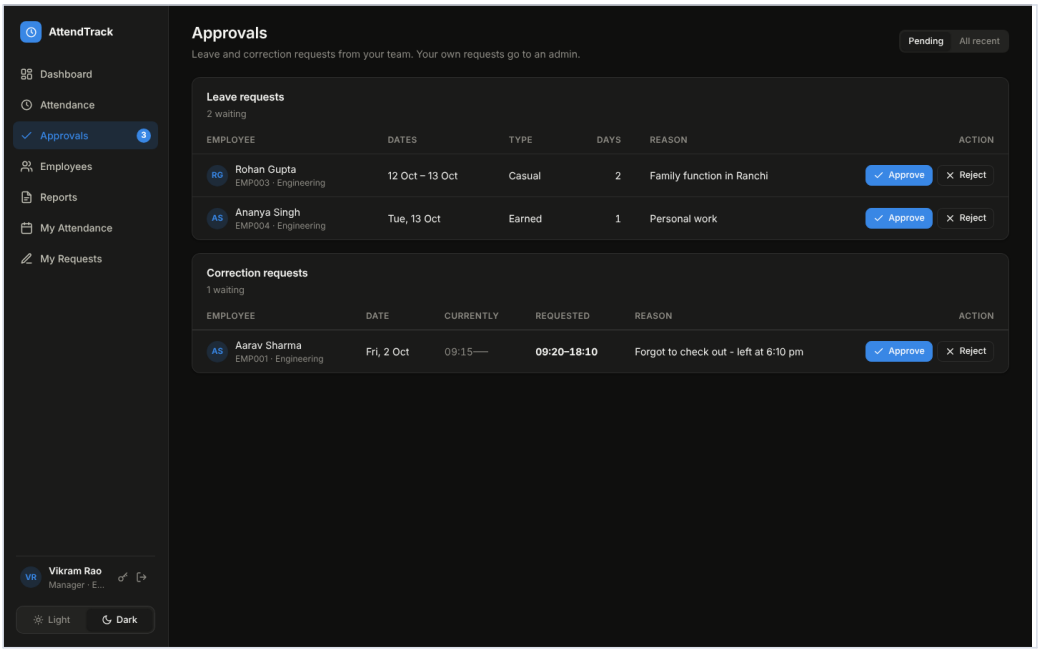
5.2 Attendance page



Attendance page: mark check-in / check-out for anyone

- Shows every employee for the selected **date** (today by default).
- Use **Check in** / **Check out** buttons to mark attendance for someone (for example, a person without a phone).
- Pick an older date to fix a record. Click the **pencil** icon to set the status (Present, Late, Half day, On leave, Absent), times and a note.
- The **bin** icon clears a record.
- Search by name or code, and filter by department or status.
- The **Note / verified** column shows how the check-in was done, for example *Self · IP 192.168.1.20 · GPS 42 m · QR*.

5.3 Approvals

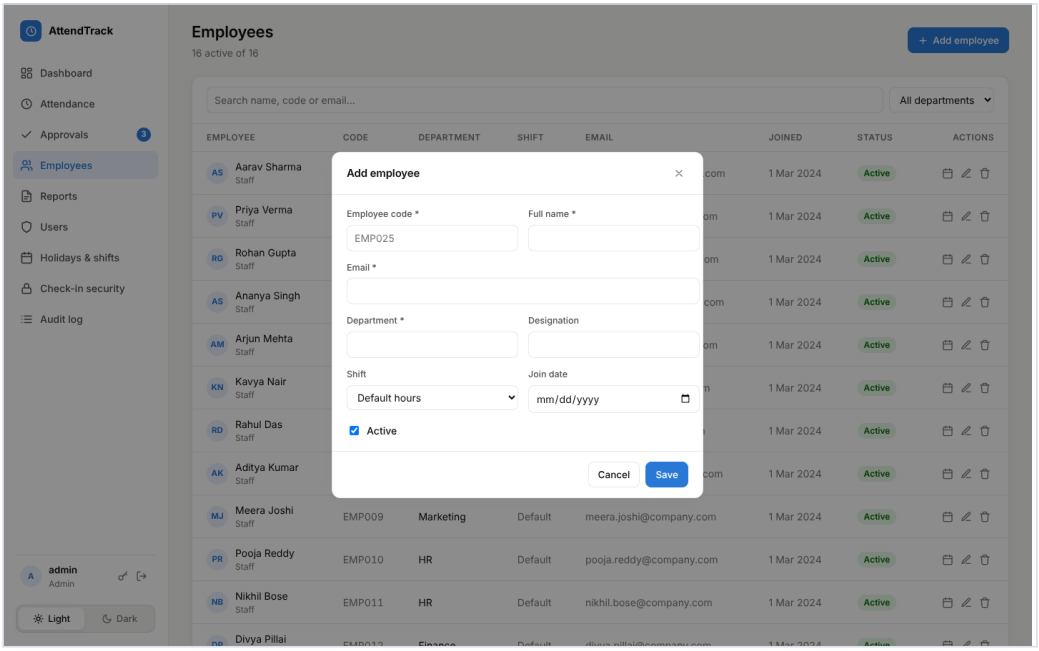


Approvals page (dark mode)

1. The red number on **Approvals** in the sidebar shows how many requests are waiting.
2. Open **Approvals**. You see **Leave requests** and **Correction requests**.
3. Check the dates, days, reason and the employee's current record.
4. Click **Approve**, or click **Reject** and type a reason (a reason is required for reject).

> **Remember:** Nobody can approve their own request. A manager's own leave must be approved by the admin.

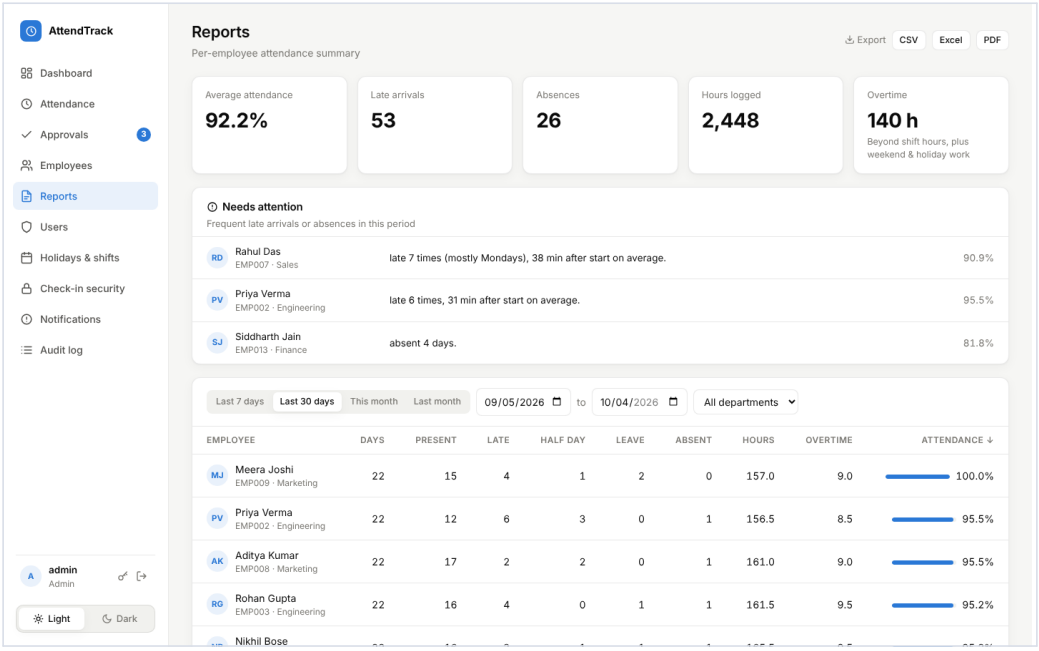
5.4 Employees



Add employee form

- **Add employee**: employee code (e.g. EMP025), full name, email, department, designation, **shift**, join date and Active.
- **Pencil** = edit, **bin** = delete, **calendar** = that person's month calendar and history.
- Untick **Active** for someone who left the company. Their old records stay.

5.5 Reports



Reports with Needs attention and Export buttons

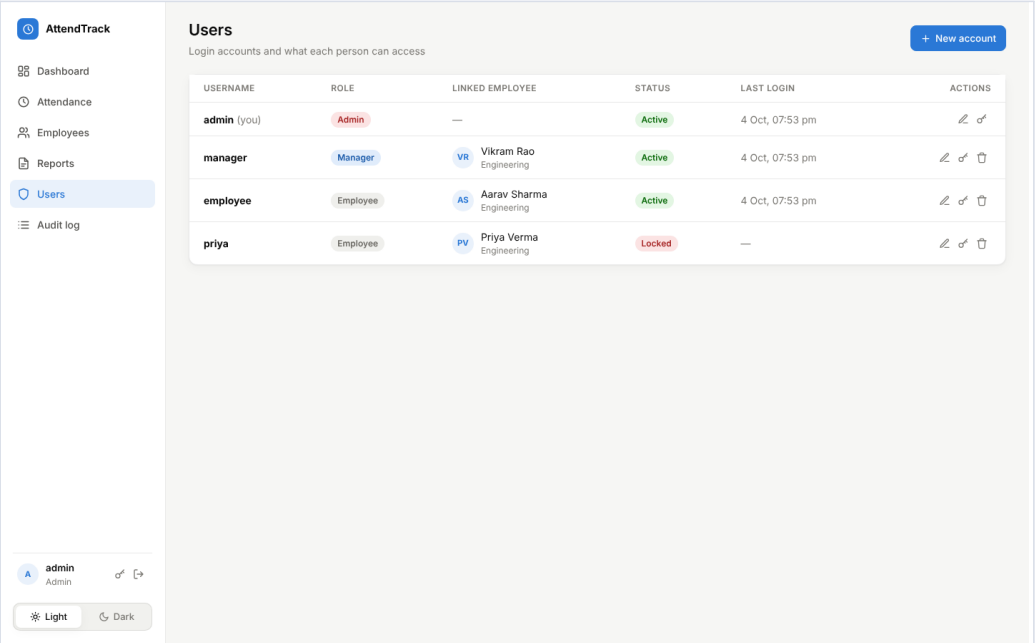
1 Choose a **Period** or set **From** and **To** dates. Pick a department if you want.

2. The table shows for each person: present, late, half day, leave, absent, hours, overtime and attendance %. Click a column heading to sort.
3. **Needs attention** lists people who were late or absent 3 or more times, for example "*late 7 times (mostly Mondays)*".
4. Click **CSV**, **Excel** or **PDF** next to **Export** to download the report.

6. Admin guide

Log in as **admin**. The admin has every manager page for **all** departments, plus five more pages.

6.1 Users (login accounts)



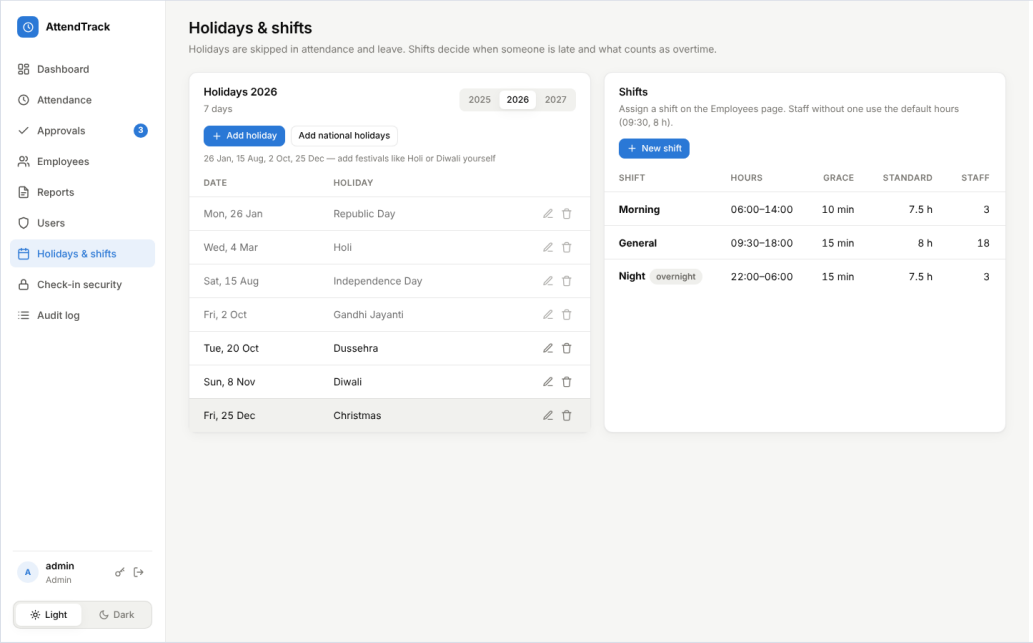
Users page

An **employee** record is a person. A **user** is a login account. One employee can have one account.

1. Click **New account**.
2. Type the **Username** and a password, choose the **Role** (Admin, Manager or Employee) and choose the **Linked employee**. A manager's department comes from the linked employee.
3. Save. Give the username and password to the person.

- **Reset password**: sets a new password for someone who forgot theirs.
- **Locked**: the account had 5 wrong passwords. It unlocks after 15 minutes, or when you reset it.
- Make an account **Disabled** to stop someone logging in without deleting it.

6.2 Holidays & shifts



Holidays & shifts page

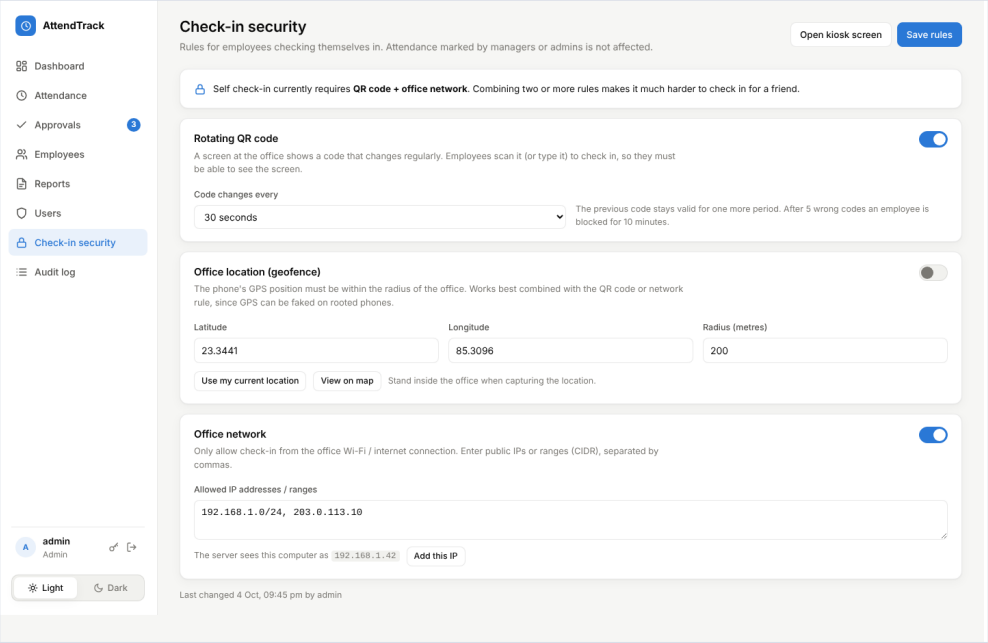
Holidays

- **Add national holidays** adds 26 Jan, 15 Aug, 2 Oct and 25 Dec for the chosen year.
- **Add holiday**: add festivals yourself (Holi, Diwali, Eid...) because their dates change every year.
- On a holiday nobody is marked absent, leave does not use the balance, and reports skip the day.

Shifts

- Three shifts are created at the start: Morning 06:00-14:00, General 09:30-18:00, Night 22:00-06:00.
- **New shift**: give a name, start, end, grace minutes and standard hours.
- Give each employee a shift on the **Employees** page. "Late" is checked against that shift.
- Night shift people can check out after midnight; hours are still correct.
- **Overtime** = hours worked beyond the shift's standard hours. Every hour on a weekend or holiday is overtime.

6.3 Check-in security



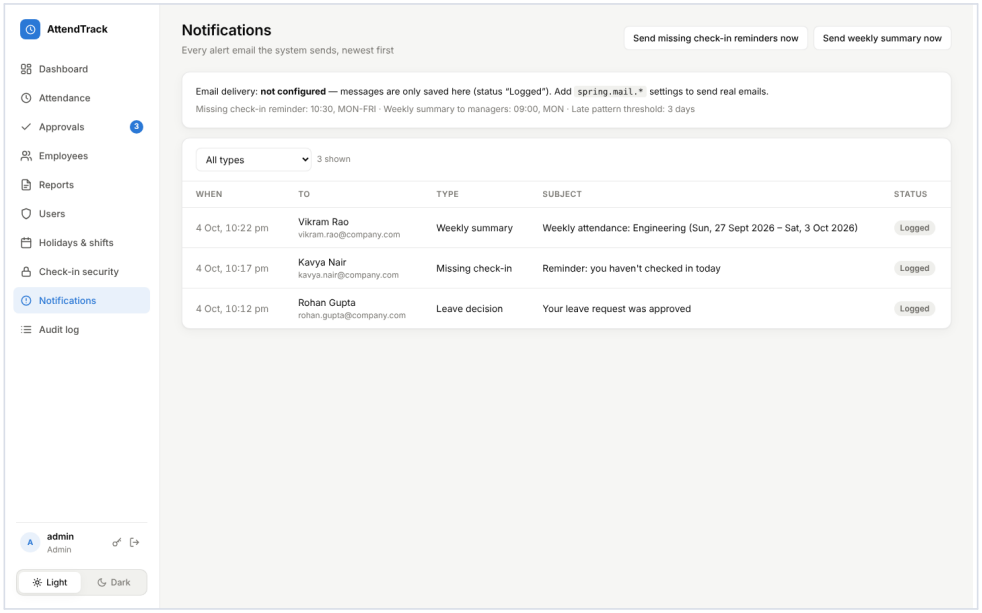
Check-in security page

These rules stop "buddy punching" (a friend checking in for someone). They apply only when an employee checks **themselves** in. Turn on any mix of the three switches, then click **Save rules**.

Rule	How it works	How to set it
Rotating QR code	Employee must enter the code shown on the office kiosk screen. The code changes every 30 seconds.	Turn on, choose how often the code changes, click **Open kiosk screen** on the office computer.
Office location (geofence)	Phone GPS must be within the radius of the office.	Stand in the office, click **Use my current location**, set the radius in metres (e.g. 200).
Office network	Check-in allowed only from the office internet (IP address).	Click **Add this IP** while on office Wi-Fi, or type IP ranges like 192.168.1.0/24.

> **Remember:** Every blocked attempt is saved in the Audit log as **Check-in rejected**. Attendance that a manager or admin marks is not affected by these rules.

6.4 Notifications (email alerts)

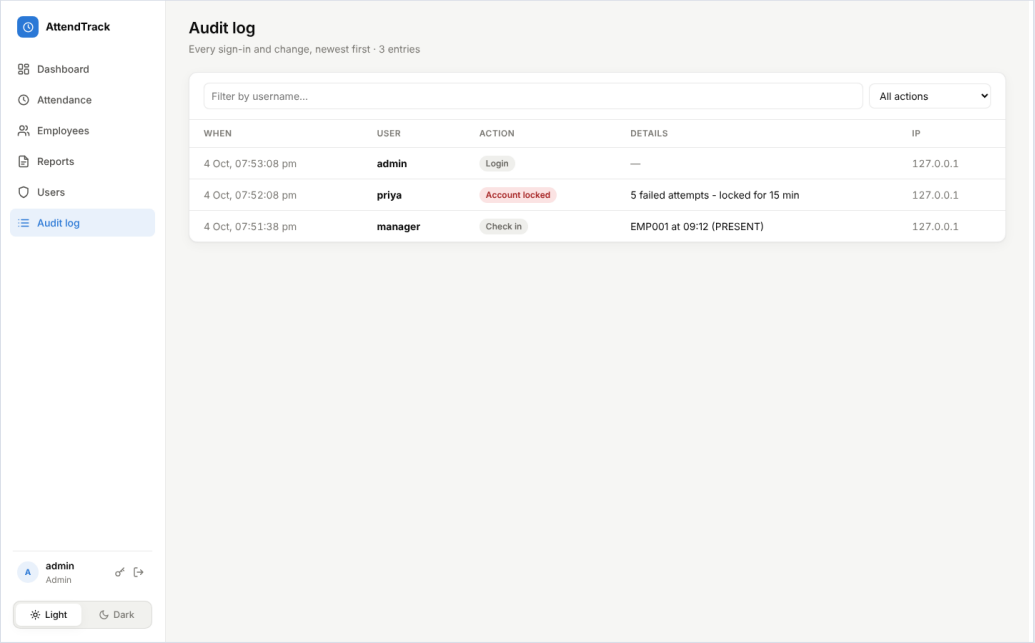


Notifications page

Alert	When	Sent to
Missing check-in reminder	Working days at 10:30	Each employee who has not checked in
Weekly summary	Mondays at 09:00	Each manager (their department) and the admin email
Leave / correction decisions	When a request is made or decided	Manager (new request), employee (decision)

Use **Send missing check-in reminders now** and **Send weekly summary now** to test. If email (SMTP) is not set up, messages are only saved on this page with status **Logged**. To send real emails, add Gmail SMTP settings (an App Password) in `application.properties` or in the `.env` file for Docker.

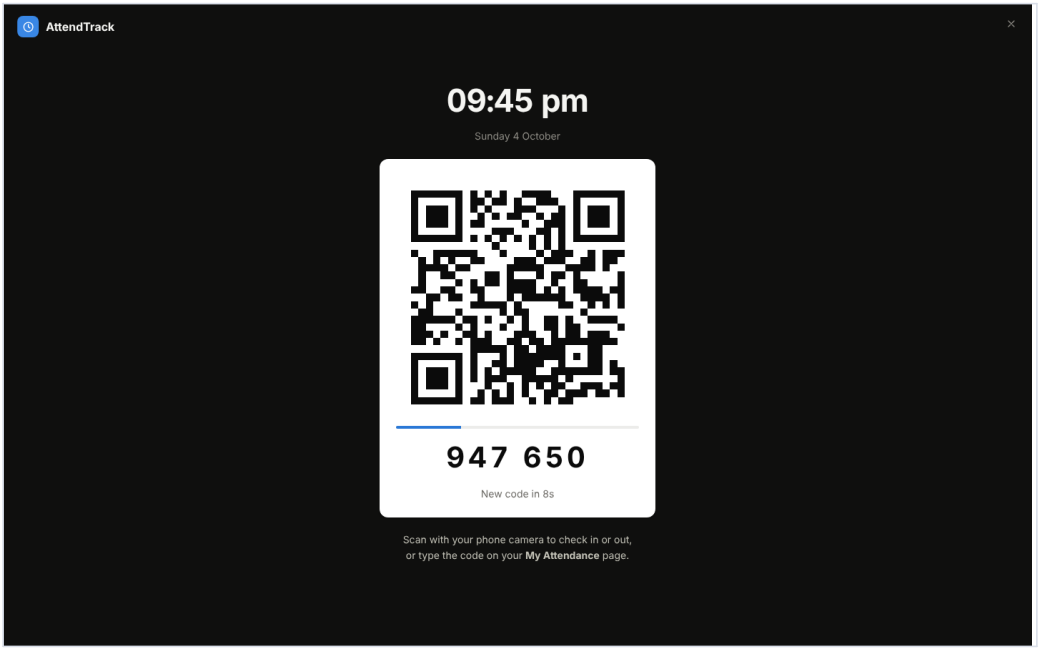
6.5 Audit log



Audit log

A record of **who did what and when**: logins, failed logins, lockouts, adding or deleting employees, approvals, rejected check-ins, settings changes. Each line has the user, time, action, details and IP address. Filter by action or username. Nobody can edit or delete these entries from the app.

7. QR code check-in (kiosk)

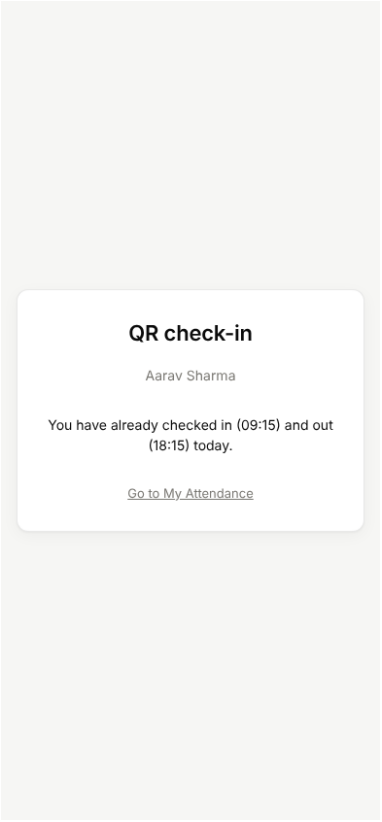


Kiosk screen on the office monitor

Setting up the kiosk (admin, one time)

1. Admin opens **Check-in security**, turns on **Rotating QR code** and clicks **Save rules**.
2. On a tablet or monitor near the office entrance, log in as admin or manager and click **Open kiosk screen** (or open `/kiosk`).
3. The screen shows a big QR code and a 6-digit code. Leave it open all day. Use the X button to exit.

Checking in with the QR code (employee, daily)



QR check-in on a phone

Way 1 - scan with phone

Open the phone camera and scan the QR code on the kiosk. A page opens. Log in if asked. You are checked in and the page shows the check-in time.

Way 2 - type the code

Open **My Attendance** on any device, click **Check in** and type the 6-digit code from the kiosk.

Why this is safe: the code changes every 30 seconds. A photo sent on WhatsApp stops working very fast, so only a person standing in the office can use it. After 5 wrong codes the person is blocked for 10 minutes.

Testing the phone camera at home

Phones allow the camera and GPS only on **https** pages. Start the frontend in phone mode:

```
cd frontend
npm run dev:phone
```

It shows an address like `https://192.168.x.x:5173`. Open it on a phone connected to the **same Wi-Fi**, accept the certificate warning, log in as employee and scan the kiosk QR on your PC. If Windows Firewall asks, allow Node.js on private networks.

8. Daily work - quick examples

Situation	What to do
Employee arrives at office	My Attendance > Check in (enter kiosk code if asked).
Employee goes home	My Attendance > Check out.
Employee forgot to check out	My Requests > Request correction > manager approves in Approvals.
Employee needs 2 days off	My Requests > Apply for leave > manager approves.

Situation	What to do
Someone has no phone	Manager opens Attendance and clicks Check in next to their name.
A new person joins	Admin: Employees > Add employee, then Users > New account linked to them.
Someone leaves the company	Employees > edit > untick Active; Users > disable the account.
Diwali holiday	Admin: Holidays & shifts > Add holiday.
Month-end report for HR	Reports > choose dates > Export Excel or PDF.
Someone forgot their password	Admin: Users > Reset password.
Check who changed a record	Admin: Audit log, filter by action or username.

9. Running with Docker (optional)

Docker runs the software the way it runs on a real server: **MySQL** database + backend + website (nginx). You only need **Docker Desktop**; Java, Maven and Node are not needed.

1. Install Docker Desktop and start it.
2. In PowerShell, go to the project folder and copy the settings file: `Copy-Item .env.example .env`
3. Open `.env` in Notepad and replace every **CHANGE_ME** (database passwords, JWT_SECRET, ADMIN_PASSWORD).
4. Run `docker compose up -d --build`. The first time takes a few minutes.
5. Open **http://localhost** and log in as **admin** with the ADMIN_PASSWORD you set.

Task	Command
See if it is running	`docker compose ps`
See backend messages	`docker compose logs -f backend`
Stop (data is kept)	`docker compose down`
Stop and delete all data	`docker compose down -v`

To put the software on the internet with your own domain and HTTPS, follow **DEPLOYMENT.md** in the project folder.

10. Problems and solutions

Problem	Reason	Solution
`mvn` is not recognized	Maven not installed or not in PATH	Install Maven, add its `bin` folder to PATH, open a new PowerShell.
"Port 8080 was already in use"	An old backend is still running	Use `start-dev.ps1` (it stops the old one), or close the old window.
Login shows "Request failed (404)"	Old backend version running, or backend not started	Stop all backend windows and start again. Wait for "Started AttendanceApplication".
Login shows network error	Backend is not running	Start the backend first.
"Account locked"	5 wrong passwords	Wait 15 minutes, or admin resets the password.
Check-in rejected	A security rule failed (code, GPS or network)	Enter the current kiosk code, allow location, use office Wi-Fi. Admin can see the reason in Audit log.
Phone camera does not open	Page is not https	Use `npm run dev:phone`.

Problem	Reason	Solution
Emails not received	SMTP not set up	Normal for the demo; see the Notifications page. Add SMTP settings for real emails.
"Running scripts is disabled"	Windows script policy	`powershell -ExecutionPolicy Bypass -File .\start-dev.ps1`
Want fresh demo data	-	Stop backend, delete `backend\data`, start again.